



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Introductions, Project, and SolidWorks Extended

Attendance



TURTLE MI

Lab Availability



TURTLE

Texas A&M University Robotics Team and Leadership Experience



Mission:

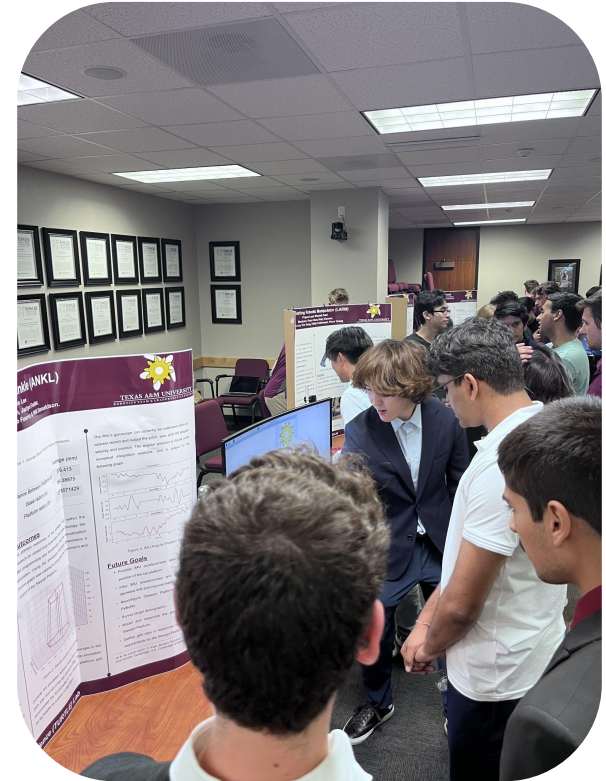
To develop the next generation of leaders through hands on robotics projects, competitions, and community service opportunities.

The oldest and largest active robotics org at TAMU

Founded in 2013

450+ Active members

18 Student Led R&D Projects





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What is Mechanical Incubator?

It is a sequel to Hatchling, focused on the robotic application of Ansys FEA, DFMA, GD&T, Electromechanical Design, and more

What Incubator really is



Hatchling introduces you to robotics and teaches you to be a robotics hobbyist. You learned the foundations and tools necessary to design, program, and build a robot.

Incubator teaches you how to be an engineer. You will leave knowing how to predict outcomes and quantify design decisions at a high undergraduate / low grad school level



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Overview

Recap of Hatchling LOs



- **SolidWorks (CAD) Competency :**

- Design custom parts and modify assemblies
- Reading and designing around manufacturing constraints

- **Electronics**

- Evaluating hardware specifications and datasheets
- Circuit design and implementation

- **Programming**

- Utilize Git/GitHub
- Control hardware via a microcontroller
- Python and C++

- **Manufacturing**

- Soldering and crimping
- Additive manufacturing

- **Problem Solving and Critical Thinking**

- Teams go through the design process from concept creation to testing

Make sure to
freshen up
on these!



MI Schedule



Week 1: Introductions, Project, and SolidWorks Revisited

Week 2: Mechanical Design & Design Requirements

Week 3: Electromechanical Design

Week 4: Materials, Manufacturing, and DFM

Week 5: Statics, Solids, and Technical Interview Concepts

Week 6: FEA I

Week 7: FEA II

Week 8: Vibrations

Week 9: Competition week!

Slides are posted on the website.

Disclaimer



We are cramming multiple 200, 300, 400, and graduate level department courses and their prerequisite into 8 weeks of 2-hour lectures

Engineering requires math. You will use it a lot in Incubator

Treat the slides as reference documents and don't hesitate to ask any questions

Doing some math



MI Directors are volunteers. What we get out of this is the ability to meet y'all and help you become the most knowledgeable version of yourself. Unfortunately **time** is the limiting factor.

Meeting hours = 8 weeks * 2 hours = 16 hours + lab times*

Tip: To get the most out of this opportunity, dedicate an extra 1-2 hours a week to learn more about the topic.

Ohh and listen to the old phrase “Ask questions”

Another Note: TURTLE



- Our capacity for MI is 30 people
 - Priority goes to new members
 - Do **NOT** expect spots to be available for repeating members next semester
- Advanced Projects are not guaranteed for MI graduates
 - Every Advanced Project lead determines their own team
 - Selection is based on the needs of the project
 - Most will look for the MI project on your resume
 - Do NOT lie about MI
 - A lot will check if you finished a MI robot
 - **If you are interested in a certain project, contact the project lead and learn more about what skills they are looking for**

General Lab Rules



- An officer must be present in the lab at all times while the lab is in use.
- Any behavior deemed disrespectful or inappropriate can result in disciplinary action or a request to vacate
- During a project's reserved lab time, the project lead may ask anyone to leave the lab for any reason.
- Project tasks always have priority over personal or school projects when it comes to lab tools and space.
- **Misuse or unsafe lab practice will result in the revoking of privileges.**
- If you need training or supervision for a tool and cannot receive it from those present, send an @Officer in the
- If at any time, common sense would defy any of the rules detailed in this document, ask an officer for help.

General Lab Rules



While in the lab, your safety is first and foremost your own responsibility. You are surrounded by resources and experienced people who can help you use them. If you are at any point unsure of what you are doing, ask. If you feel unsafe with a tool, stop what you are doing immediately and do not resume work until you have received clarification.

Tools Requiring Training:

- Angle Grinder
- Dremel
- Drill Press
- Hand Drill
- Impact Driver
- Jig Saw
- Power Supply
- Printer (3D)
- Soldering Iron

Tools that require Supervision:

- Angle Grinder
- Dremel

Safety Gear

- PPE is required for all tools listed in the training section excluding the 3D printers. Proper PPE includes: closed toe shoes, bottoms that cover to the lower leg, a sleeved shirt, and safety glasses. All loose items and hair must also be properly secured up and out of the way.

Trainings

- Any individual may be trained by an officer. This training is tool specific and is not all encompassing.



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Officer Introductions

Drew Wheaton - MI Director

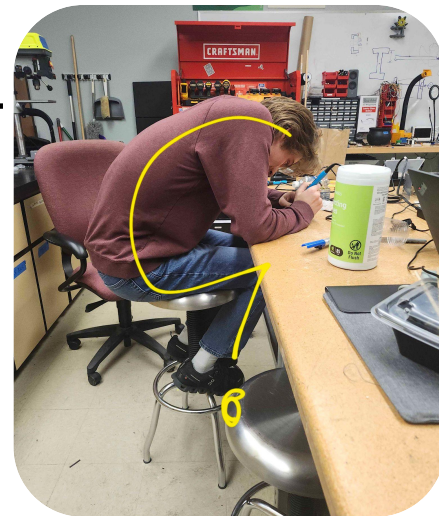


Email: dwwheaton01@tamu.edu

Discord: loafofbread77



- MEEN 2027
- Meeting Days: M&T
- Experience:
 - Former Hatchling
 - DRON & POBS
- Fun facts:
 - Can do the splits
 - Pirate font handwriting
 - Read 11 books in a month



Julia Sopala - Development VP



Email: jusopala18@tamu.edu

Discord: juli.a9847



- BMEN 2027
- Meeting Days: TBD
- Experience:
 - OLSN (Prosthetic hand project)
 - R&D Engineering Internship
 - Biomedical Research Lab Experience
 - Hatchling Instructor
 - Former Hatchling
- Fun facts:
 - I want to be fluent in 4 languages
 - I'm 3/5th of the way there
 - I want to learn how to surf

Ethan Real - MI Instructor



Email:ethanreal05@tamu.edu

Discord: ethan_real05



- MEEN 2027
- Meeting Days: Mondays
- Experience:
 - Turtle
 - Former Hatchling Member
 - BEEST
 - Interned @ Sulzer
- Fun facts:
 - I'm a Fraternal Twin
 - I love to go fishing whenever I can

McKenzie McCain - MI Instructor



McKenzie McCain

Email: kemping01@tamu.edu

Discord: kemping01



- ITDE 2027
 - Interdisciplinary Engineering (Biomechanics & Robotics Track)
- Meeting Days:
 - TBD
- Experience
 - TURTLE
 - Former Hatchling
 - OLSN
 - Interned @ RAD Lab
- Fun Facts
 - I know the 100 digits of pi
 - My initials are $E=(Mc)^2$



Click to add title



Khanathan Vo

Email: Khanh0315@tamu.edu

Discord: M1necrftpr0



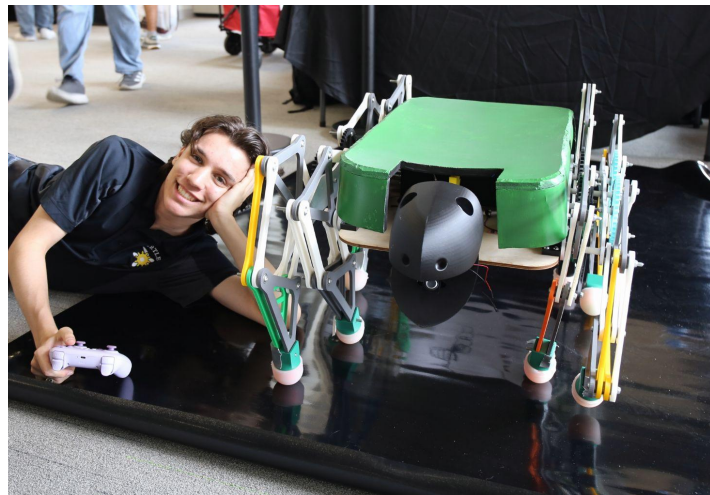
- MEEN 2028
- Meeting Days:
 - Tuesday
- Experience
 - TURTLE
 - BLNC
 - POBS
 - Hatchling Fall 2025 winner
 - HERC Lab OSL
 - Doordash Driver
- Fun Facts
 - I like skating and running and volleyball (larp)
 - I like sleeping in the lab or scc

Lorenzo Hidalgo - MI Instructor

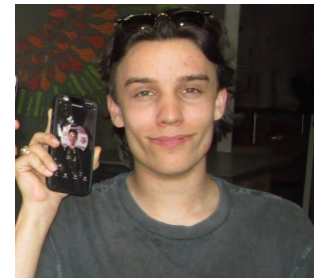


Email: lhidalgo@tamu.edu

Discord: darthzo



- MEEN 2027
- Meeting Days
 - Mondays and Tuesdays (depending on schedule 😊)
- Experience:
 - TURTLE
 - BEEST
 - SHDR
 - OLSN
- Fun facts
 - I've had white hair since I was born
 - I've had two family disappearances



William Shan - MI Instructor



William Shan

Email: williamshan698@tamu.edu

Discord: [williamthepig1](#)



- MEEN 2028
- Meeting Days: Mondays and potentially Fridays
- Experience:
 - TURTLE
 - Former Hatchling
 - CMBT
 - Dig 'Em Aggies
 - Aggies Create
 - Interned @ Flowserve
- Fun facts:
 - TURTLE Smash Bros Mega Champion
 - basketball



Andrew Sun - MI Instructor



Email: andrewsun.15@tamu.edu

Discord: goatedogger



- MMET 2029
- Meeting Days: Monday sometimes
- Experience:
 - Former Hatchling
 - CMBT
 - Aggies Create
- Fun facts:
 - I'm friends with the goat
 - Tennis
 - Basketball
 - Severance & AOT are my fav show of all time





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Icebreakers!

Name and Fact



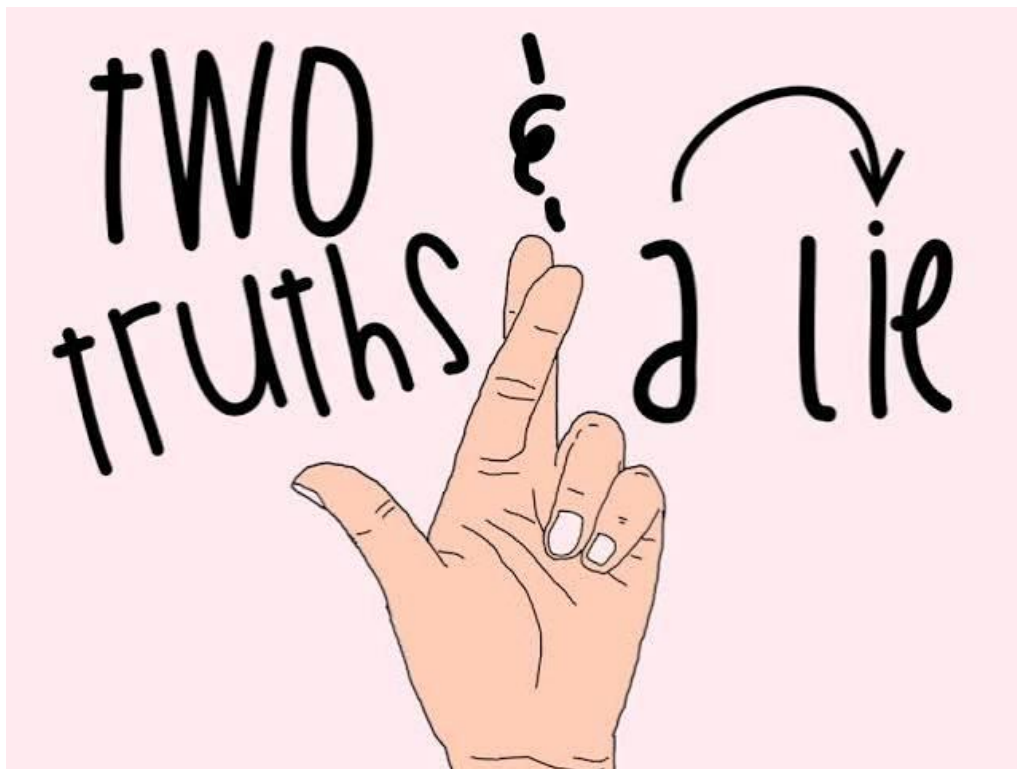
Talker: Say your name, year, major, and one ~~fun~~ fact about yourself

Everybody: Raise your hand if the fact applies to you



After everyone has finished, pair up with someone who had a similar fact

2 Truths and a Lie



After everyone has finished, pair up with someone across the room and repeat.



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Project Intro: BattleBots

Design, build, and test a Plastic Antweight combat robot

Big Picture Expectations



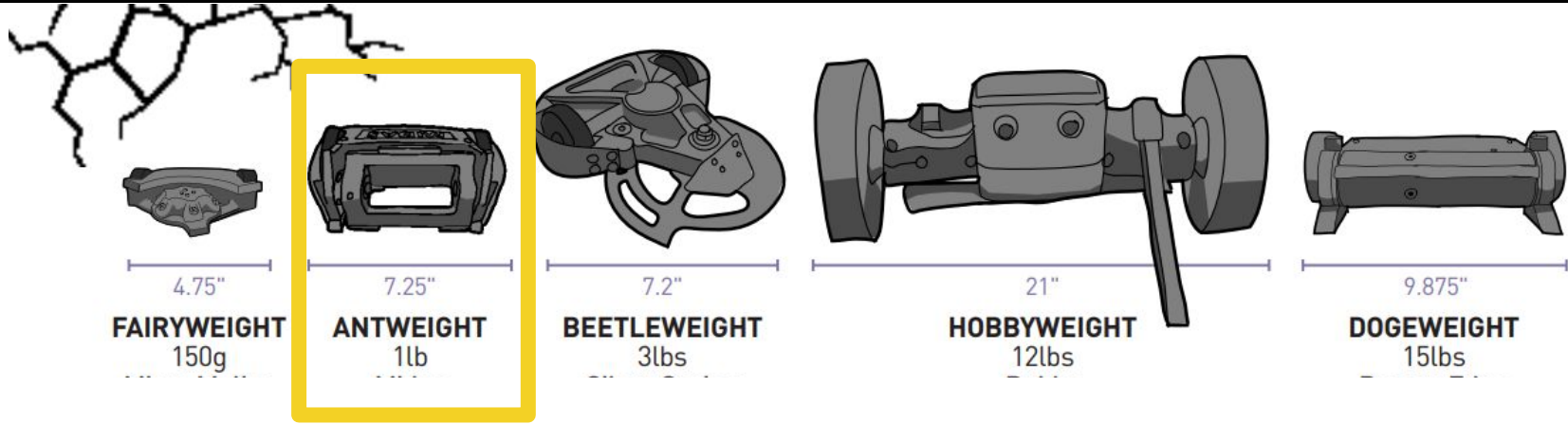
- Work in small engineering teams (teams of 3)
- Design a competition-ready combat robot
- Apply concepts from each MI workshop directly to the robot
- Manufacture and assemble the robot throughout the semester
- Test, iterate, and improve the design
- Compete against other MI teams, and potential school-wide competitions at the end of the program

What is Combat Robotics?



- Head-to-head robot competition
- Robots that use weapons, mobility, and control to defeat an opponent
- Designs must satisfy strict weight, size, safety, and construction requirements
- Success requires balancing performance, reliability, manufacturability, and strategy

Combat Robot Weight Classes



We are building the Plastic Antweight! (1 lb maximum)

Plastic-focused construction rules make design, weight management, and manufacturing especially important.

How a Match Works



This tournament is held under [SPARC](#) procedures, each match will be 3 minutes

Before the match

- Place the robot in its starting position with its drive wheels pointed away from people and other robots, then power receiver or robot
- Once both robots are in their starting squares, the arena will be closed
- Drivers may perform a brief weapon/drive test for up to 20 seconds
- The referee will ask both drivers if they are ready, then start the match.

Knockouts/immobilization

- A robot must demonstrate controlled movement. If immobilized, a 10 second countdown begins, failure to move before zero results in a knockout
- If the robot's battery becomes exposed, the referee will stop the match and award the victory to the opponent
- Pinning: Any robot performing a "stationary pin" is limited to 5 seconds, an "active pin" is limited to 10 seconds, anytime after that, the robot in control must release the opposing robot
- Matches will be paused to separate robots if they happen to get stuck together. Robots will have 10 seconds to separate. If unable to separate an unstick will be called by the referee

How a Match Works



Judges' Decision

- If neither robot is knocked out or taps out before the match ends, the match goes to the judges
- Judges evaluate the robots based on Damage, Control, and Aggression Criteria (found in SPARC Judging Guidelines v1.3)

Tap-outs

- At any time during a match the robot operator may choose to tap out by verbally notifying the ref

Safety Procedure

- Stop fighting immediately when instructed by the referee
- Once stopped, robots will be deactivated and no work may be performed on them until officials determine the next step
- The match may resume from where it was stopped if the safety issue is resolved; otherwise, the match will be judged up to the point of stoppage
- Failure to stop fighting when instructed can result in disqualification.

Damage, Control, and Aggression Criteria



If the match goes to the judges, and the robot with the most points wins. There will be a 3 judges and each will have pools of points to give each bot.

Here's what the judges are looking for:

- **Damage** – 6 points: How much you impair your opponent's robot
- **Aggression** – 5 points: How consistently you attack and pursue your opponent
- **Control** – 6 points: How well you dictate the fight and control your opponent

What should our robot be designed to do well?

- **Deal Damage + Stay Aggressive + Control The Fight**

Example



Here's an example of how scoring would work with Robot A vs. Robot B

Category	Judge 1	Judge 2	Judge 3
Damage (6)	A: 3 – B: 3	A: 4 – B: 2	A: 3 – B: 3
Aggression (5)	A: 3 – B: 2	A: 3 – B: 2	A: 2 – B: 3
Control (6)	A: 3 – B: 3	A: 3 – B: 3	A: 3 – B: 3
Judge's Total	A: 9 – B: 8	A: 10 – B: 7	A: 8 – B: 9

Combat Robot Safety



Along with following the general lab rules, you must also:

- Never activate a weapon outside approved testing conditions
- Treat every robot as potentially active
- Remove/disable power before working on the robot
- Secure robots during weapon testing
- Follow battery charging/storage procedures
- Follow instructor/officer directions during testing
- Stop immediately if a robot behaves unexpectedly

There will be a quiz before the tournament to check that you understand the SPARC rules!! Please read them they are not long.

Additional rules and regulations can be found [here](#) (SPARC Documents)

Robot Construction Rules



SPARC has construction specifications. The key ones for your project are sections 1–6, 8, and 16.3 (Plant section)

Some key requirements:

- 1 lb Maximum
- indicator that robot is powered
- weapon lock

Common Combat Robot Archetypes



Vertical Spinner:
Weapon on robot that spins
on the vertical plane to
attack

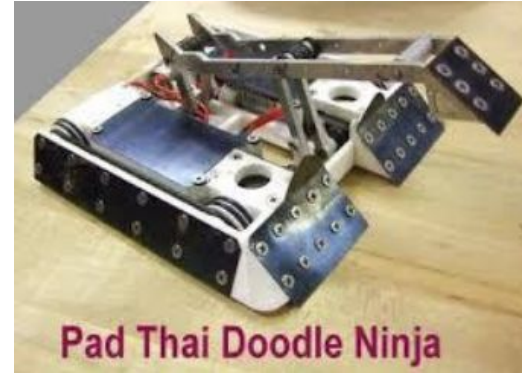


Horizontal Spinner:
Weapon on robot between the
bottom and top of chassis that
spins parallel to the ground to
attack

Common Combat Robot Archetypes



Drum Spinner:
Fast spinning cylindrical
drum at front to tear
opponents apart



Flipper Robots:
Robots that focus on
launching opponents in the
air

Common Combat Robot Archetypes



Control Bot:
Robots that focus on seizing, lifting, or pinning to control opposing robots



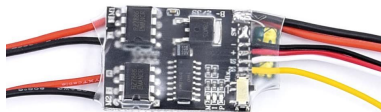
Hammer Bot:
Robots with a weapon that swings over the top down to hit the opposing bot

[Great Place to Start for more info`](#)

Available Components



- PLA+
- 2S LiPo
- brushed motors (4 max)
- servo motor or brushless motor
- dual brushed ESC
- brushless ESC
- FlySky Tx&Rx
- wire, solder, heat shrink, connectors
- screws, nuts, heat-set inserts
- If you need more, ask and we may be able to get it!



Project Timeline



Week 1 (9/14):

- Start forming teams
- Research PLANTs and brainstorm

Week 2 (9/21):

- Finalize teams
- Continue Brainstorming

Week 3 (9/28):

- Finalize robot concept
- Select key components
- Begin CAD
- *Milestone 1* (Bot concept and part selection)

Week 4 (10/5):

- Continue CAD

Week 5 (10/12):

- Finish CAD
- Begin building
- *Milestone 2* (CAD review and printing plan)

Week 6 (10/19):

- Continue Building

Week 7 (10/26):

- Continue Building
- *Milestone 3* (Final check)

Week 8 (11/2):

- Continue Building

Week 9 (11/9):

- Finish robot
- Compete!

Lab Time Availability



- We will be scheduling two 1 hour dedicated lab times this week
- Please fill out your availability so we can schedule times that you can come
- You don't need to come to each time, but it's dedicated for you to work on your project.
- Likely on W, R, or F afternoon/evening





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SolidWorks Extended

Always remember Design Intent

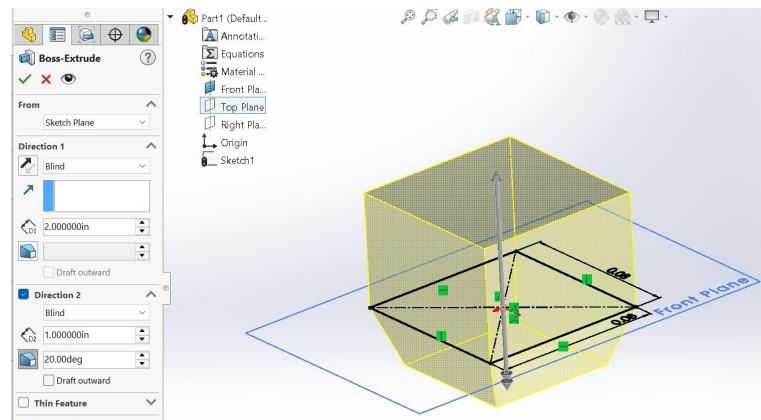
The Basic CAD Journey



1. Create/Select a Plane (A face is a plane)
2. Create a Sketch
 - a. Draw general shape w.r.t the origin
 - b. Add relations
 - c. Dimension the rest from largest to smallest

**Always ensure
you are fully
defined before
step 3**

3. Turn Sketch into a 3D Feature
4. Save
5. Repeat

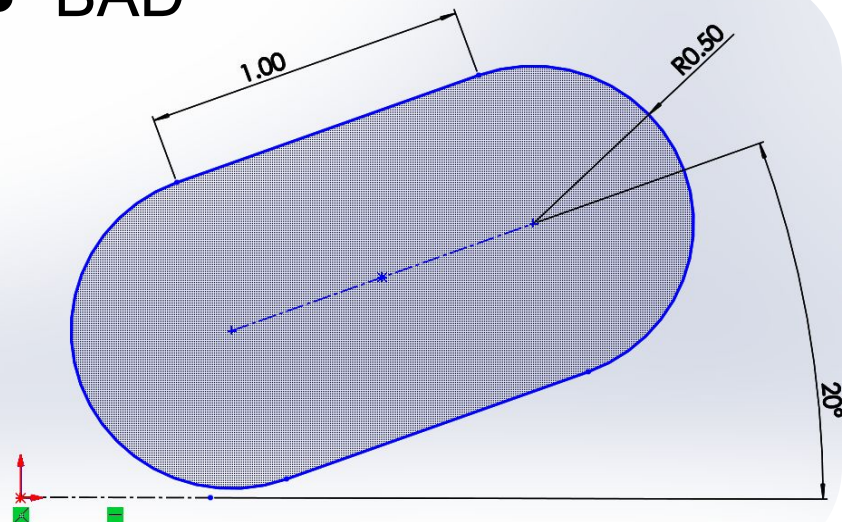


Is my sketch fully defined?



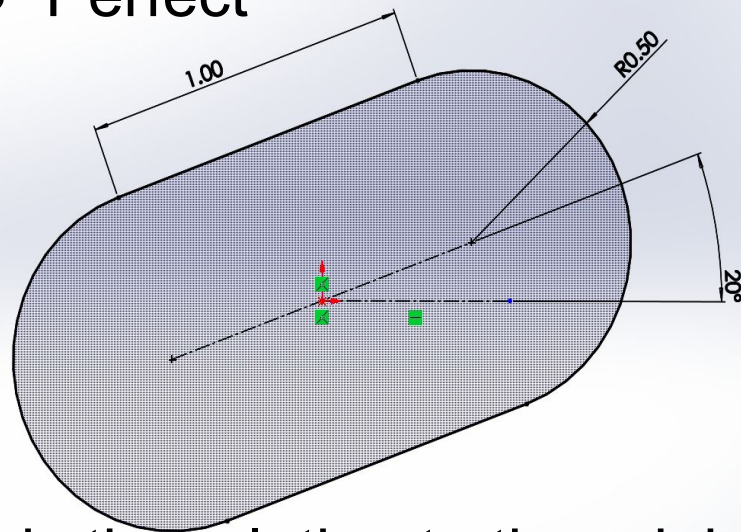
Under defined:

- Blue Lines
- Multiple Potential Solutions
- BAD



Fully defined:

- Black Lines
- One Solution
- Perfect



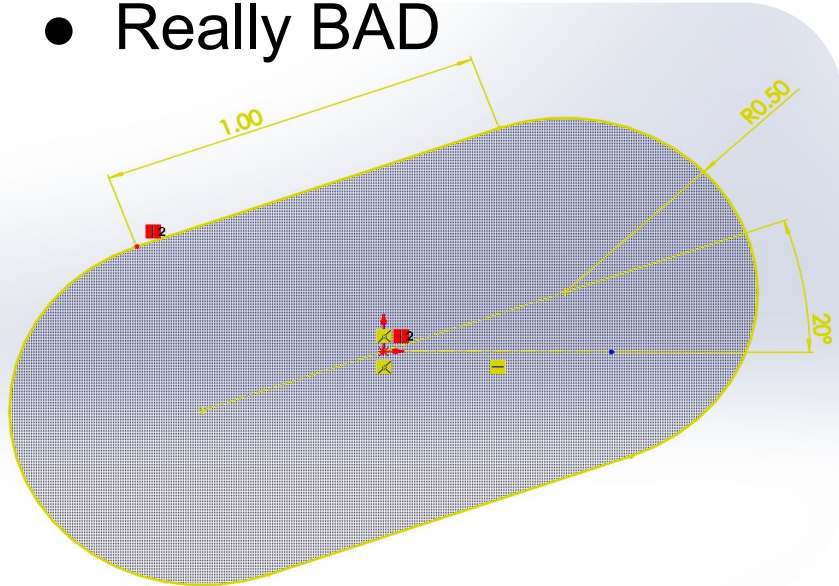
The difference between these is the relation to the origin.

Other Defined States



Over defined:

- Yellow/Red Lines
- Zero Potential Solutions
- Really BAD



Construction (Centerlines)
lines:

- Reference Geometry indicated as a dashed line
- Has no direct impact on potential solutions
- Very useful for defining curves



Origin Selection



1. Identify Planes/Axis of Symmetry
 - a. Primary → Entire Body
 - b. Secondary → Features that can be mirrored or pattern
2. Pick the point that has the most intersections of step 1.
 - b. Prioritize the primary symmetric geometry
3. If step 2 has multiple points, chose one with a prominent feature

Want a Unique Reference Plane?



It is often advantageous to use a custom reference plane when designing parts. You can do this through the “Reference Geometry” tool.

A plane can be uniquely defined by:

- Three non-collinear points
- A point and a line not on that point
- Two distinct intersecting lines
- Two separate parallel lines
- A flat face



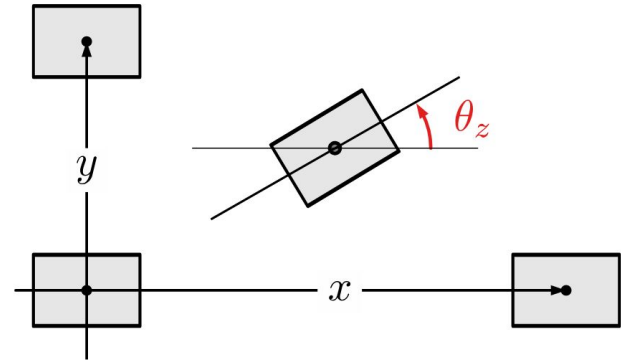
Sketches



Sketches are 2D “drawings”.

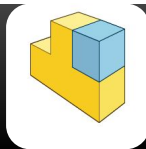
A plane has 3 Degrees of Freedom:

- X (Translational)
- Y (Translational)
- $\theta(z)$ (Rotational)



engineeringstatics.org

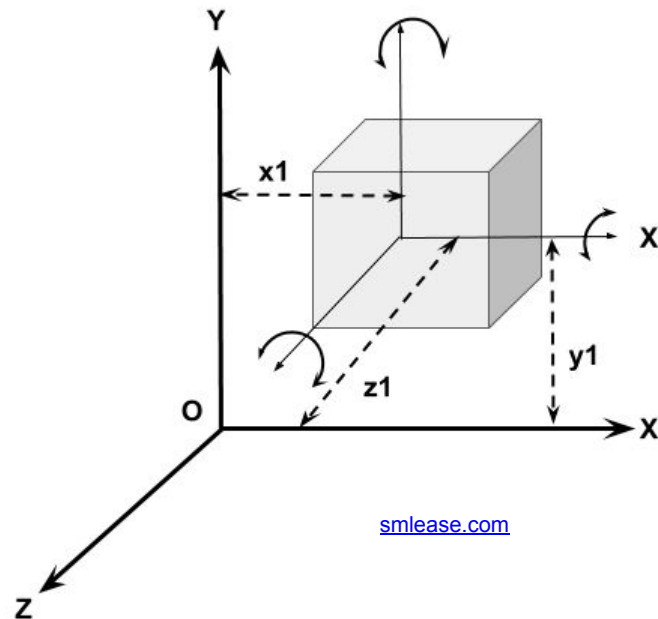
An ideal sketch will have one enclosed region and fully defined (cannot move) in space.



Assemblies are 3D

3D has 6 Degrees of Freedom

- X (Translational)
- Y (Translational)
- Z (Translational)
- $\theta(x)$ (Rotational)
- $\theta(y)$ (Rotational)
- $\theta(z)$ (Rotational)

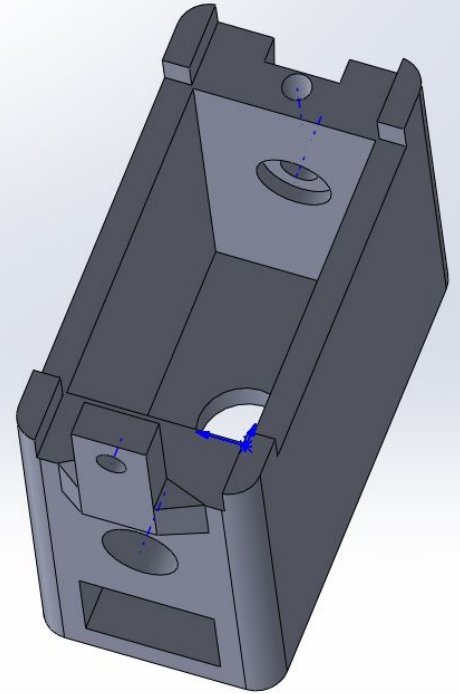
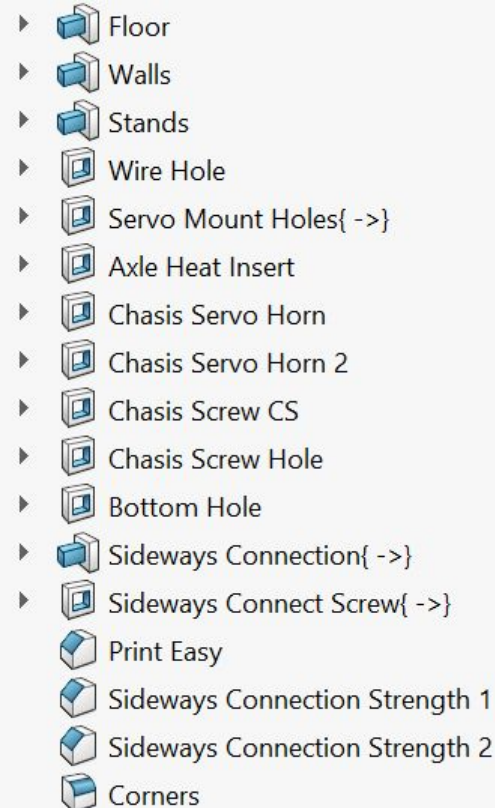


Moving mechanism may result in under defined assemblies

Resilient Modeling Strategy - FTM



1. Add Material
2. Subtract Material
3. Special Operations
 - a. Shell, Patterns, Mirror
4. Finishing Operations
 - b. Chamfer, fillets
5. Repeat Loops
 - c. if necessary



Renaming features was a personal preference

SolidWorks Folders - FTM



Necessity when organizing large assembly feature-trees

Please use for your fasteners

To use:

1. Right Click feature-tree
2. Select one of the two options
3. Click and drag components



You can suppress every component in one click

Sub-Assemblies - (FTM)



- You can add assemblies to assemblies
 - You cannot add an assembly to itself
- Mates within sub-assembly do **NOT** calculate
 - This is **essential** for large (100+) component assemblies
 - Think modularity. Every module should be a sub-assembly
- Flexible sub-assemblies do calculate internal mates
 - Only use if you absolutely must 

COTS Work Smarter NOT Harder



Most item suppliers will provide a CAD file. Use them

- SolidWorks accepts most CAD file formats

Common suppliers:

- [McMaster-Carr](#): Fasteners, Power Transmission, etc
- [Misumi](#): Industrial components
- [Adafruit](#): Electronics
- A lot more

McMASTER-CARR

Lg., mm	Threading	Thread Spacing	Head Dia., mm	HT, mm	Drive Size, mm	Tensile Strength, psi	Specifications Met	Pkg. Qty.	Pkg.
18-8 Stainless Steel									
2	Fully Threaded	Coarse	2.6	1.4	1.3	70,000	DIN 912, ISO 4762	5	91292A328 \$16.30
3	Fully Threaded	Coarse	2.6	1.4	1.3	70,000	DIN 912, ISO 4762	5	91292A329 12.43
4	Fully Threaded	Coarse	2.6	1.4	1.3	70,000	DIN 912, ISO 4762	5	91292A330 13.08
5	Fully Threaded	Coarse	2.6	1.4	1.3	70,000	DIN 912, ISO 4762	5	91292A331 13.20
6	Fully Threaded	Coarse	2.6	1.4	1.3	70,000	DIN 912, ISO 4762	5	91292A332 13.29

*We have M3, M2.5, or M2 screws in the lab

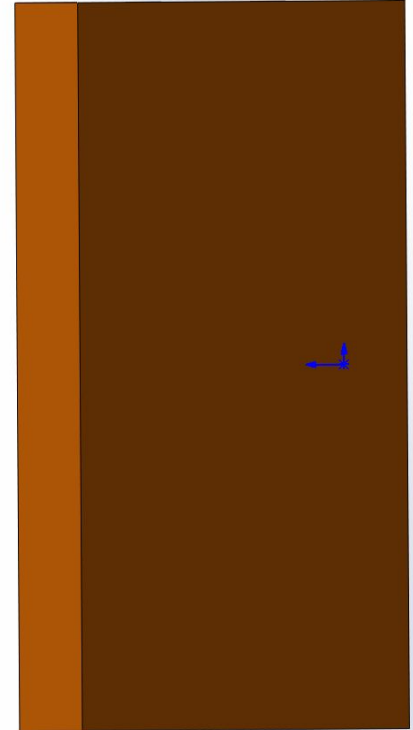
Part Appearance



1. Select the four color ball in the View (Heads Up) bar
2. Select intended scope (Part, Face)
3. Select a color or appearance

* There is no functional purpose outside of renders

** Typically used to identify between parts in an assembly



Measure Tool

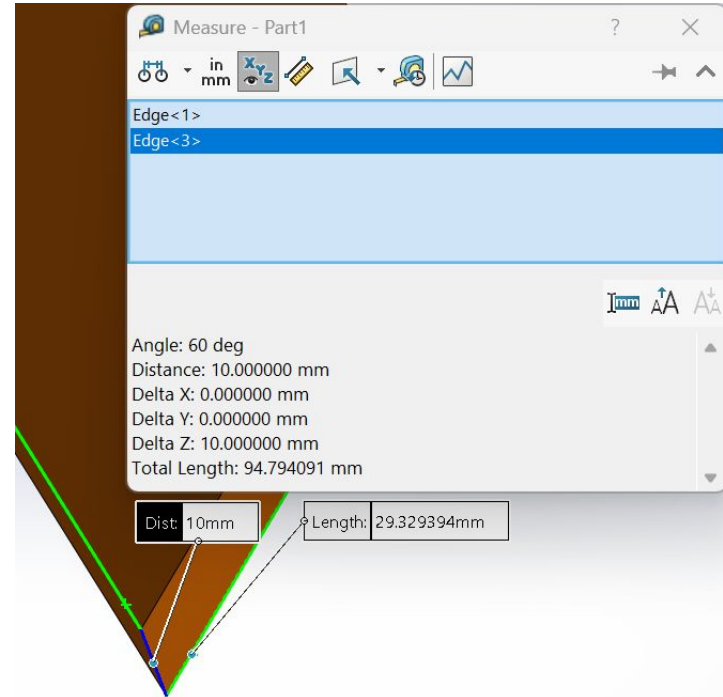


1. Click “Measure” within “Evaluate” tab
2. Select geometry to measure

Mostly used for:

- Determining various distances
- Angle between faces/edges
- Length of edges
- Area / Perimeter of faces

*Works in parts and assemblies



Mass Properties

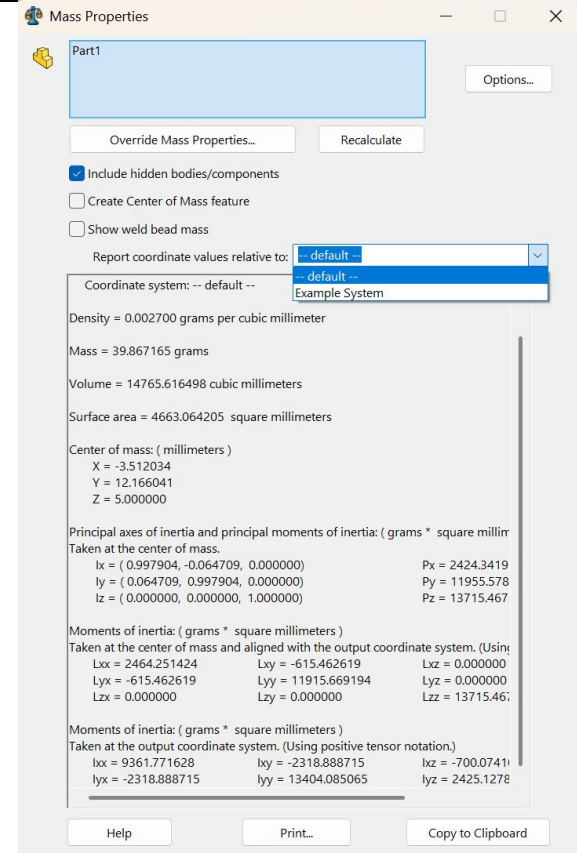


1. Click “Mass Properties” within “Evaluate” tab
2. Select coordinate system

*Make sure to select the correct coordinate system

Gives:

Mass, Volume, Surface area, CoM, Inertia Axis, and Inertia



Configurations



Used to create a multiple variations of a part or assembly within a single file

Advantages:

- Prevents modeling same thing twice
- Allows design table parametric modelling
- Reduces the number of files you need



Mcmaster M3
Screw

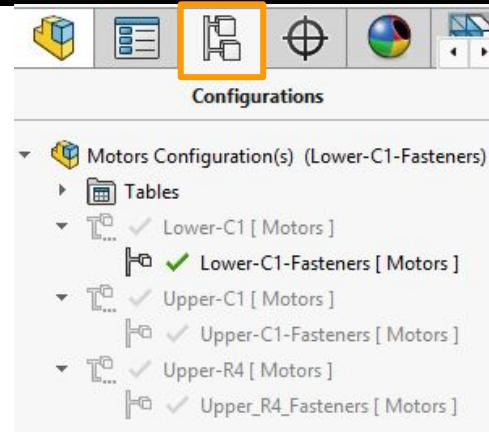
Think about screws. One CAD model works for every length variation

Using Configurations



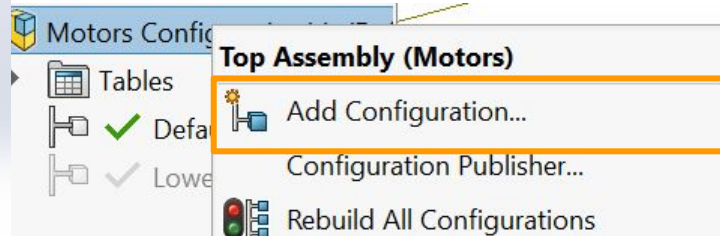
How to Use:

1. Switch to “Configurations” tab
2. Right click then “Add Configuration”
3. Double Click to switch between configurations



*Green Check is the configuration you are in

Configurations work on both parts and assemblies



Design Table



Implementation of Excel in SolidWorks for parametric modeling More powerful version of “Global Variables”

Sketch Dimensions

Design Table for: Involute Gear, Metric

	Tooth Width@Sketch2	Root Fillet@Root Fillet	Tip Radius@Tip Radius	\$STATE@Hole	HoleDia@Sketch6	SOLIDWORKS Materials:ABS PC Gear, Metric Spur, DT 2021	PressAngle_Measured@Sketch2	Outside_Diameter@Sketch2	Root_Diameter@Sketch2	Tooth_Depth@Sketch2	\$PR@UNIT_OF_MEASURE	Shaft Dia@Sketch5	Module	Pressure Angle	Tooth Form	Outside_Diameter@Sketch2	Root_Diameter@Sketch2	Pressure Angle Error, Decimal Degrees	Pressure Angle Error, Seconds
1m, 25 Deg, 10T, Standard, Center Hole	1.571	0.35	0.13	U	3	SOLIDWORKS Materials:ABS PC	25.0025	12.000	7.500	2.250	- none - 10		1.0	25	Standard	12.000	7.500	-0.00252	-9.07
1m, 25 Deg, 21T, Standard, Center Hole	1.571	0.35	0.13	U	3	SOLIDWORKS Materials:ABS PC	25.0001	23.000	18.500	2.250	- none - 14		1.0	25	Standard	23.000	18.500	-0.00007	-0.27
0.8m, 20 Deg, 24T, Stub	1.257	0.28	0.10	S	11	SOLIDWORKS Materials:6061 Al	19.9999	20.480	17.600	1.440	- none - 14		0.8	20	Stub	20.480	17.600	0.00011	0.39
0.8m, 20 Deg, 33T, Standard	1.257	0.28	0.10	S	15	SOLIDWORKS Materials:Brass	20.0002	28.000	24.400	1.800	- none - 20		0.8	20	Standard	28.000	24.400	-0.00016	-0.57
0.8m, 25 Deg, 24T, Standard, Center Hole	1.257	0.28	0.10	U	10	SOLIDWORKS Materials:AISI 304	24.9999	20.800	17.200	1.800	- none - 14		0.8	25	Standard	20.800	17.200	0.00014	0.50
0.8m, 25 Deg, 24T, Stub, Center Hole	1.257	0.28	0.10	U	10	SOLIDWORKS Materials:AISI 316 Annealed Stainl	25.0001	20.480	17.600	1.440	- none - 14		0.8	25	Stub	20.480	17.600	-0.00015	-0.53
Green = Input																			
Orange = Secondary Input																			
Yellow = Calculated																			
Driven Value for Reporting in DT																			

Configurations

Design Table - Setup



Learn more about how you can use design tables to let you build and manage variations of a part or assembly in a single file:

Quick YouTube how to [create a design table](#)
How to [create a design table blog](#)

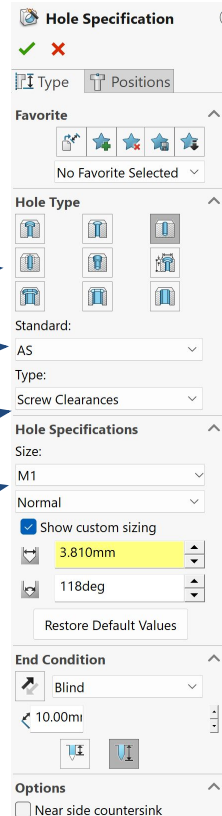
Hole Wizard



Allows for the creation of standardized holes, and is a Sketch and Feature creation tool in one.

To use:

- Click on Hole Wizard in the Features tab
- Select the hole type
- Specify the unit standard
- Type - provides standard sizing for drill bits, dowels, tap sizes, and screw clearances
- Size - includes normal and fits for screw clearances (loose, normal, close)
- Specify end conditions



Hole Wizard - Hole Types



Counterbore



Legacy Hole



Countersink



Counterbore Slot



Hole



Countersink Slot



Straight Tap



Slot



Tapered Tap

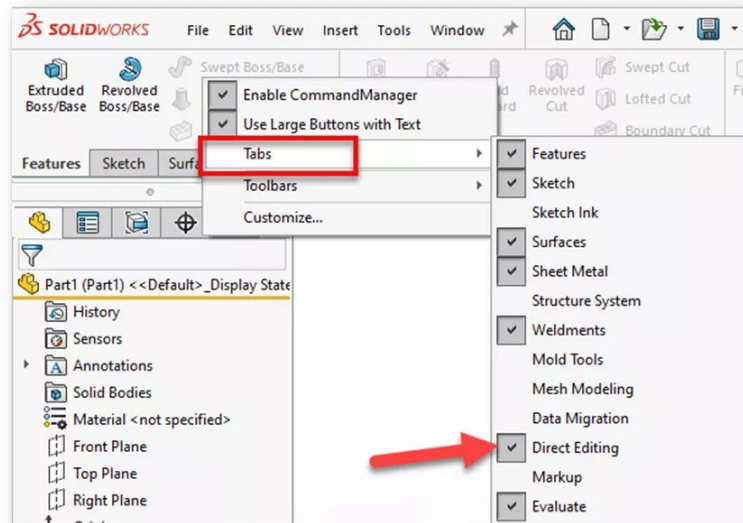
Direct Editing



In SolidWorks, Direct Editing tools provide the flexibility to modify geometry that does not contain model features, such as an imported STEP, Parasolid, or IGES file.

To turn on the Direct Editing tab, right-click anywhere on the CommandManager, go to Tabs, then select **Direct Editing**.

[Instruction Guide](#)



Direct Editing - Move Face

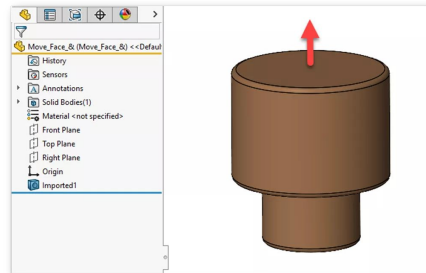
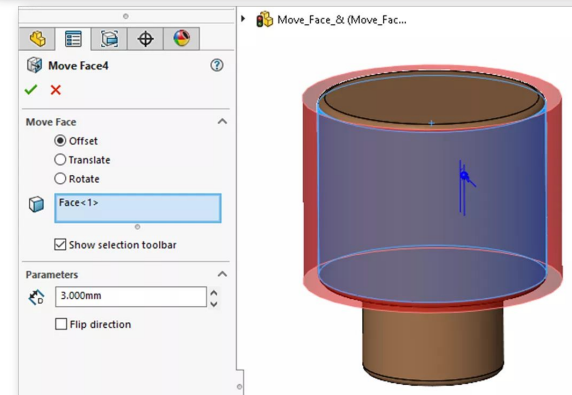
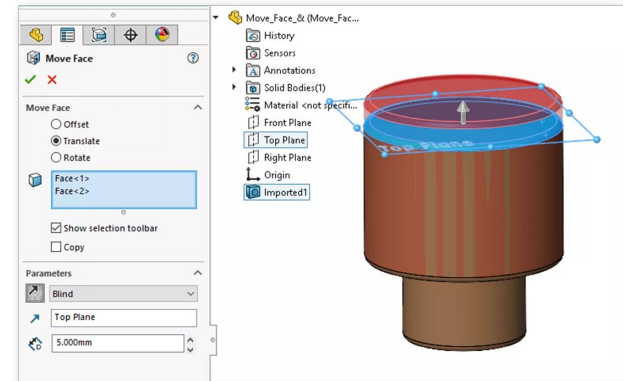


The Move Face feature can help you adjust the size of your part in different directions.

On the **Direct Editing** tab, choose **Move Face**, you could then select **Offset**, **Translate**, or **Rotate**.

Then, you can select which faces that need to be moved.

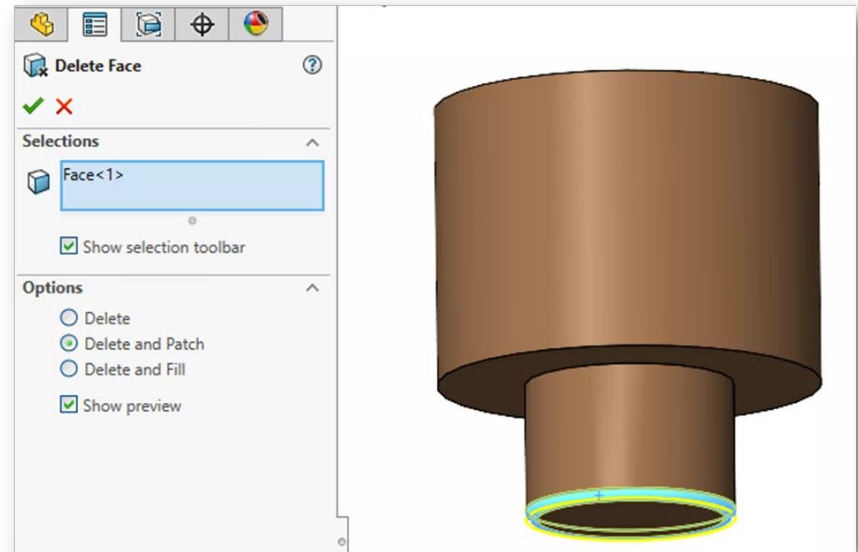
Under **Parameters**, you can choose your Move's end condition, select a plane for the direction reference, then key in your desired length and click **OK**.



Direct Editing - Delete Face



The Delete Face feature can help us remove remaining fillets and add chamfers on edges. To remove the fillet in the example to the right, you can use the **Delete Face** command on the Directing Editing tab. Under **Selections**, choose the fillet face. Under **Options**, select **Delete and Patch** and click **OK**.



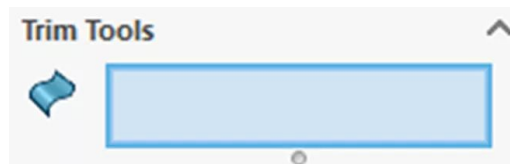
Direct Editing - Split



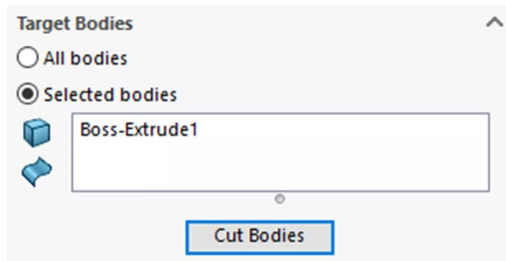
The "Split" command (Insert > Features > Split) is a great tool for breaking a solid body into multiple sections and/or deleting sections of a solid body or bodies.

The Split command is split into three primary areas:

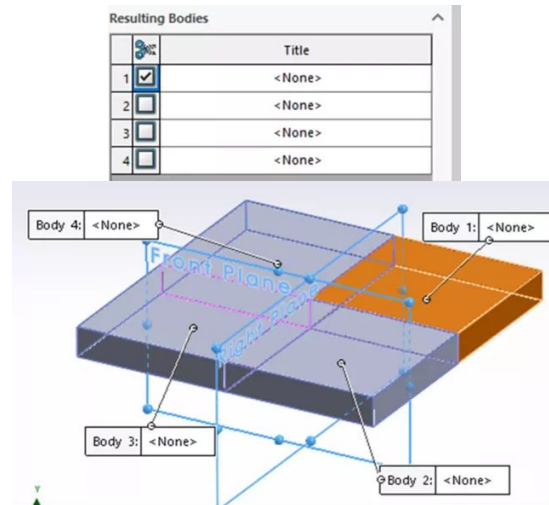
1. Trim Tools



2. Target Bodies



3. Resulting Bodies



[More step by step for split feature](#)

Direct Editing - Combine

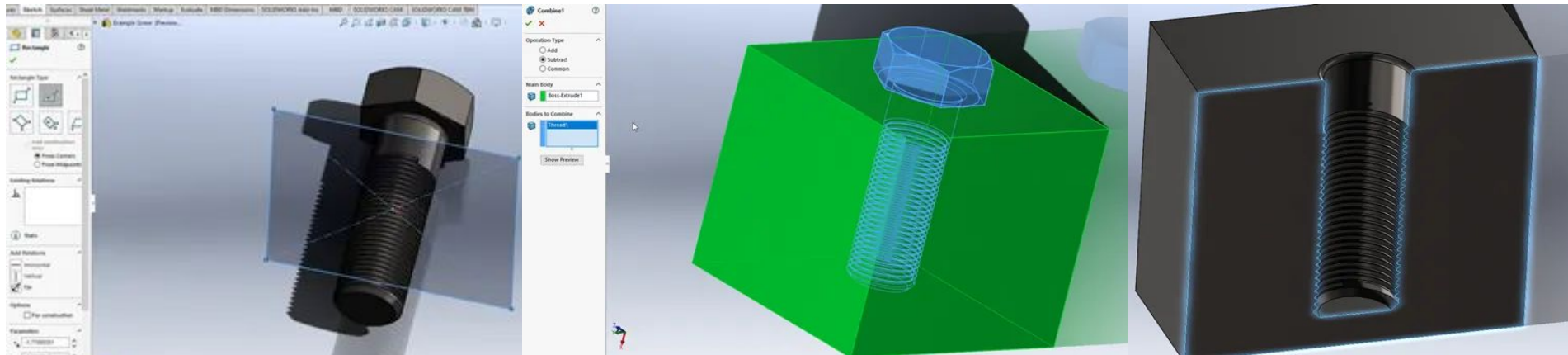


The Combine feature is a great tool used most often in surface modeling as it allows you to take molds of created parts and negative spaces for a part.

The big idea is to create a box or extrude over your base component, use the combine feature to create a negative of the part in that extruded piece.

IMPORTANT: Make sure to uncheck **Merge Results** as the default for SOLIDWORKS is to merge the bodies. By unselecting this, we ensure the parts are separated making the next command easier.

[Step by step for combining](#)





TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Program Installation yay!

Download the slides at:

turtlerobotics.org/MI

*Always recommend bringing a mouse and your laptop charger



Texas A&M students have free access to ANSYS Academic Enterprise through the Software Center. **Recommended**

<https://software.tamu.edu/>

DO NOT
INSTALL IN
ONEDRIVE

DO NOT
INSTALL IN
ONEDRIVE

Note: ANSYS is the recommended FEA Software for TURTLE Advanced Projects. Please try to utilize ANSYS, however, other FEA platforms may be accommodated.

Note 2: You must be connect to tamu_wifi or the TAMU VPN to access the license

The software can also be accessed through the Virtual Open Access Lab (VOAL) which can be used without downloading Ansys.

[Omissa Horizon Clients]

and

People who are currently installing ANSYS

<https://voal.tamu.edu/>

Cisco Secure Client (TAMU VPN)

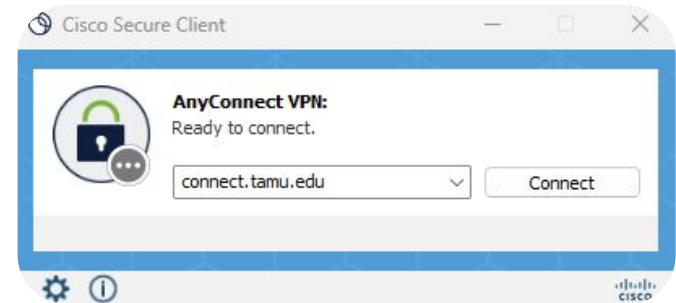


Instructions can be found at <https://tinyurl.com/56u37d8p>

Summary

1. Download Client at connect.tamu.edu
2. Set server as connect.tamu.edu
3. Connect and login with NetID

*This allows you to access Aggie Cinema off campus



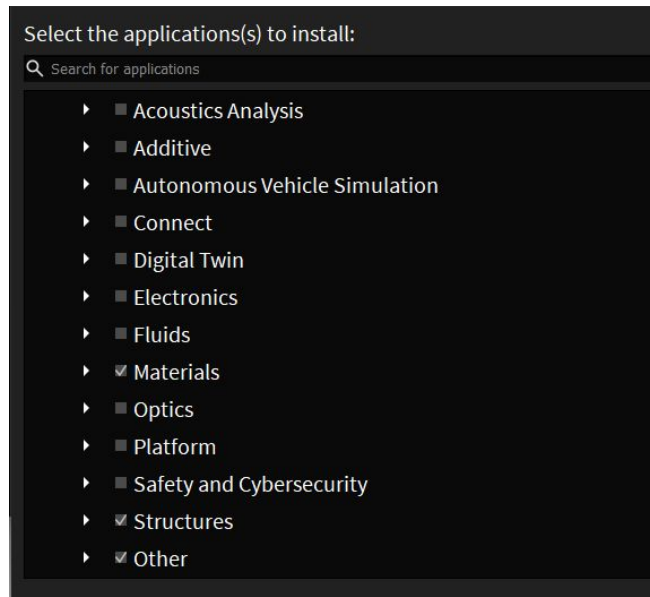
ANSYS Install



Install instructions can be found at <https://tinyurl.com/yw6wm5dc>

Summary

1. Download / run ANSYSinstaller.exe
2. Go through prompts
3. Products - defaults (~72 GB)
4. License
 - a. Installations
 - i. Port: 1055
 - ii. Server 1: coe-vtls7.engr.tamu.edu
 - b. Press Test for a 
5. Run ANSYS Workbench



For more info on which apps give you which products, consult [Ansys Install Guide](#)

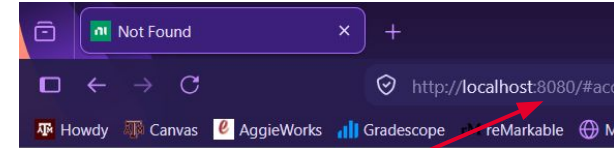
Common Errors



- ANSYS install Access Error: 404 -- Not Found Cannot locate document:

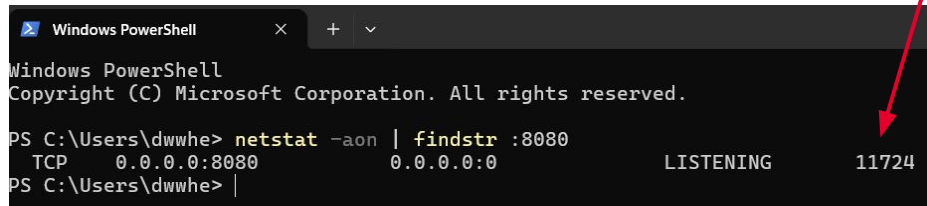
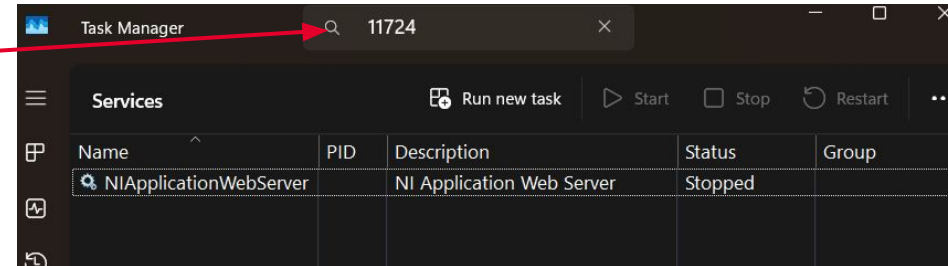
- Ensure you are running the exe as administrator

- Open PowerShell and run “netstat -aon”
 - Find the local host address under the “Local Address” column (netstat -aon | findstr :<port number>)
 - Find the corresponding PID
- Open Task Manager and the “Services” tab
 - End task on the conflicting PID



Access Error: 404 -- Not Found

Cannot locate document:





TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

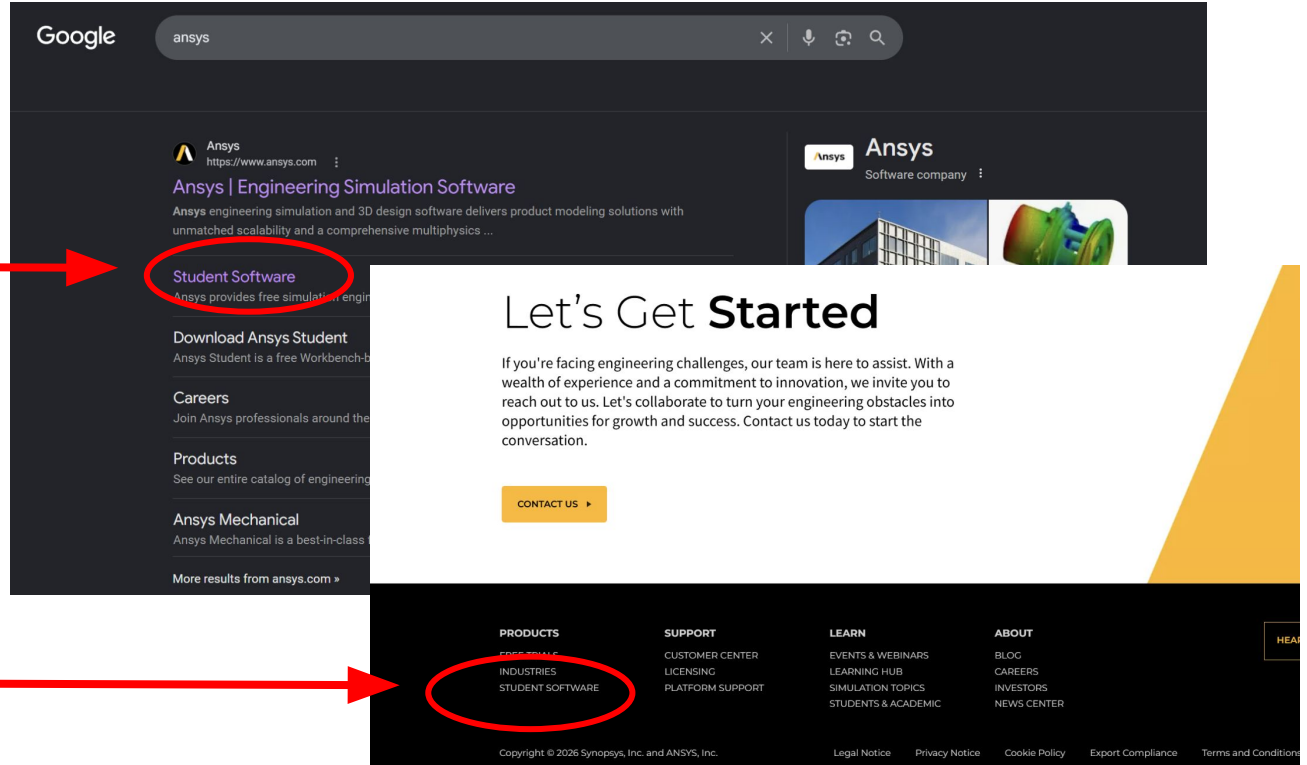
Alternative method to download ANSYS

(does not require account or VPN)

Alternative Method



Go to the ANSYS website and click on student software

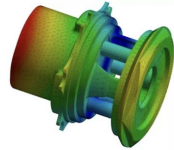
The image shows a screenshot of the ANSYS website. The top navigation bar is dark blue with the ANSYS logo on the left and a search bar on the right. Below the navigation bar, the main content area is white. On the left side of the main content area, there is a sidebar with a list of links: 'Student Software', 'Download Ansys Student', 'Careers', 'Products', 'Ansys Mechanical', and 'More results from ansys.com'. The 'Student Software' link is circled in red. A red arrow points from the text 'Go to the ANSYS website and click on student software' to this link. The main content area features a large heading 'Let's Get Started' and a paragraph of text. Below the text is a yellow button labeled 'CONTACT US'. At the bottom of the page, there is a dark blue footer with a grid of links under the headings 'PRODUCTS', 'SUPPORT', 'LEARN', and 'ABOUT'. The 'STUDENT SOFTWARE' link under the 'PRODUCTS' heading is circled in red. A red arrow points from the text 'Go to the ANSYS website and click on student software' to this link. The footer also contains copyright information and various legal notices.

Alternative Method



Free Student Software Downloads

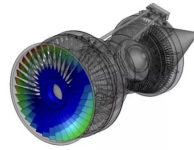
Used by millions around the world, students can take advantage of our free engineering software for homework, capstone projects and student competitions. Our renewable products can be downloaded at no cost by students across the globe and installed on any supported MS Windows 64-bit machine.



Ansys Student

Ansyes Student is our Ansys Workbench-based bundle including Ansys Mechanical, Ansys CFD, Ansys Discovery, Ansys Rocky, Ansys optiSLang, Ansys Speos, Ansys Zemax OpticStudio and so much more. Ansys Student is downloaded by hundreds of thousands of students globally every year and includes some of our most-used products commercially. Users of this product may also find value in downloading our Ansys LS-DYNA Student or Ansys Rocky Student Add-Ins product

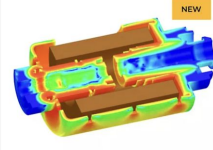
[FREE DOWNLOAD NOW ▶](#)



Ansyes LS-DYNA Student

Ansyes LS-DYNA Student has a vast array of capabilities using its explicit solver, allowing the simulation of the response of materials to short periods of severe loading and extreme deformation. It can be used by students to examine materials failure and how failure progresses through a part or system. LS-DYNA Student is useful for applications in automotive, aerospace and more.

[FREE DOWN](#)



Ansyes Electronics Desktop Student

Ansyes Electronics Desktop Student is our industry gold standard simulators for work with antenna, RF, Microwave, PCB, IC and IC package designs along with electromechanical devices such as electric motors and generators. Students will have access to Ansys HFSS, Ansys Maxwell, Ansys Q3D, and Ansys Icepak allowing design work a broad range of electrical and electro-mechanical systems.

Ansyes Student - Free Software Download

Ansyes Student offers free access to our Ansys Workbench-based bundle. This bundle includes Ansys Mechanical, Ansys CFD, Ansys Discovery, Ansys SPEOS, Ansys Zemax OpticStudio, Ansys Autodyn, Ansys optiSLang, Ansys Rocky, and much more (see "What's Included" below for a complete list). Used by students across the globe, Ansys Student can be leveraged to enhance your skill set with some of our most-used products.

From 2025R2 on, Ansys Zemax OpticStudio is now included in Ansys Student. Ansys Zemax OpticStudio Student offers free access for students to gain hands-on experience in designing and analyzing optical systems, preparing them for future careers in optics. Students will have access to an intuitive interface and powerful tools with extensive materials and optical elements libraries, enabling them to simulate real-world scenarios accurately, aiding their understanding of optics fundamentals, and enhancing their problem-solving skills.

Terms of Use: Ansys Student downloads are for educational use only and may only be used for self-learning, student instruction, student projects, and student demonstrations.

[DOWNLOAD ANSYS STUDENT 2026 R1](#)

(This license valid until 6/30/2027)

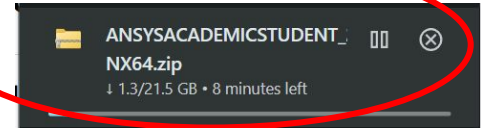
For the [free online simulation course from Cornell University](#), Ansys Student 2026 R1 is recommended.

Ansyes Rocky Add-Ins can be downloaded from [here](#).

Download
ANSYS Student

This will lead to
this page where
you can click
download

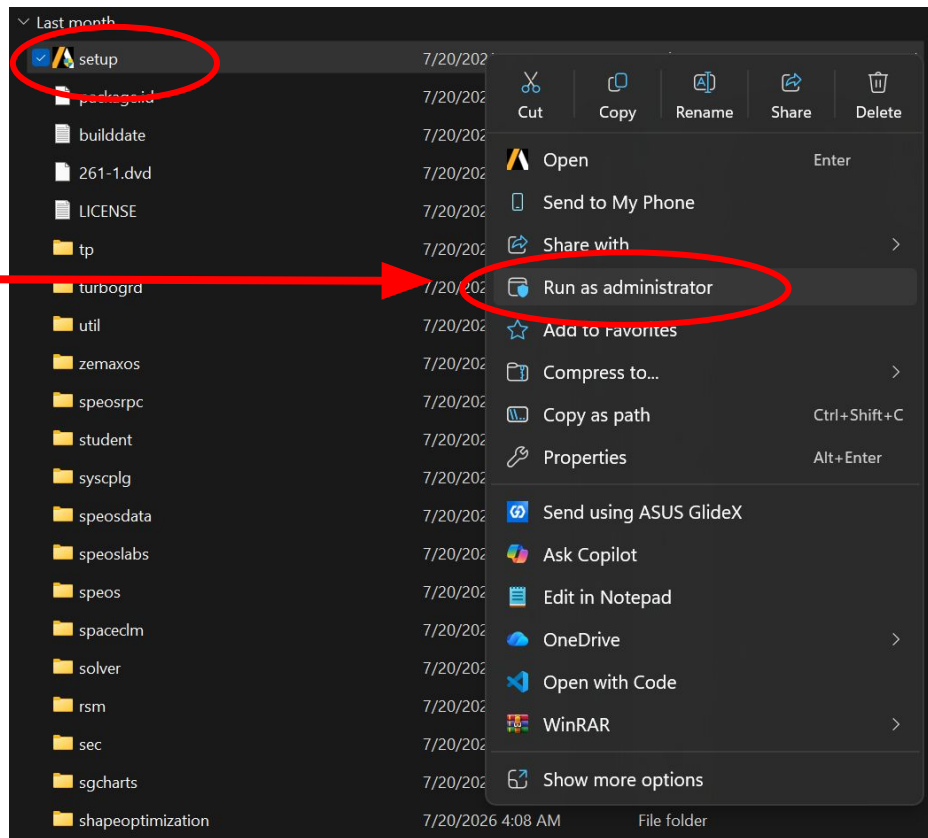
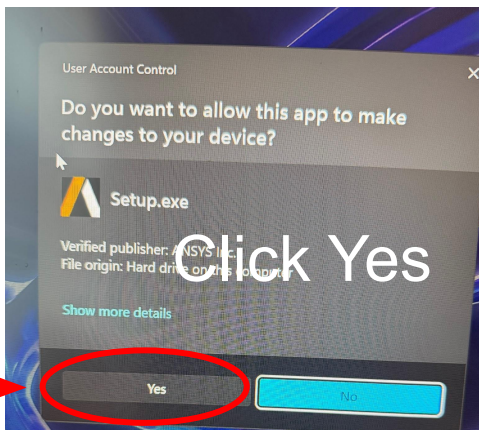
Once it is finished
downloading, Unzip this file



Alternative Method



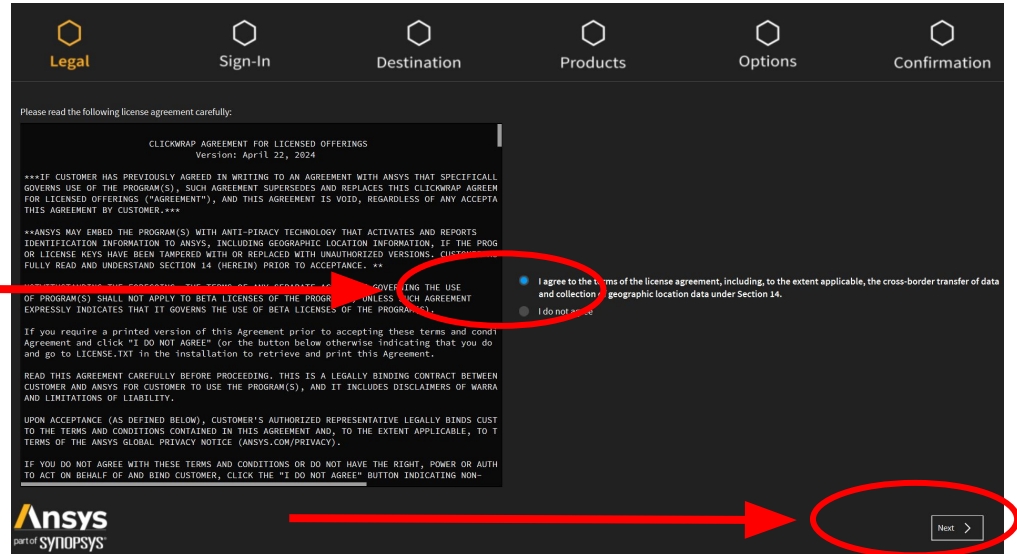
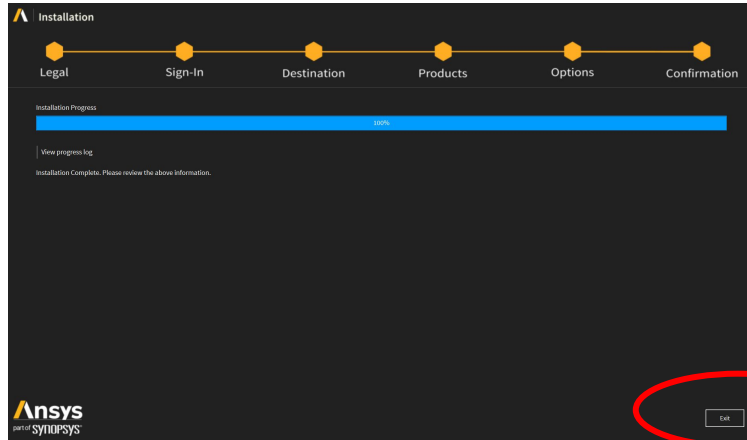
Right click the setup exe file
and run as administrator.



Alternative Method

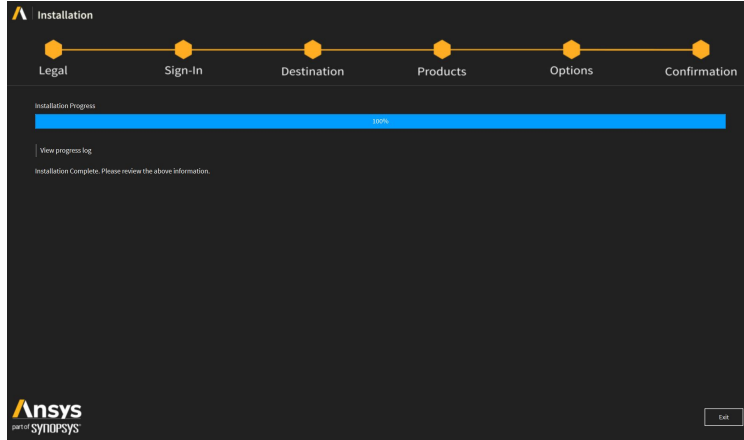


Agree to this
then press next
until the end



Exit and restart PC when done

Alternative Method Extra



Just check what version and type its asking for and go to the dot net [website](https://dotnet.microsoft.com/en-us/download/dotnet/8.0) and download and install that too (outside the scope of the tutorial)

<https://dotnet.microsoft.com/en-us/download/dotnet/8.0>

Install screen mighttttt prompt you to download dot Net 8.0 software

Download .NET 8.0

Not what you're looking for? Visit the [downloads](#) page for more options.

8.0.29 Security patch

[Release notes](#) Latest release date July 14, 2026

Build apps - SDK

SDK 8.0.423

OS	Installers	Binaries
Linux	Package manager instructions	x64 Arm64 Arm32 x64 Alpine Arm64 Alpine Arm32 Alpine
macOS	x64 Arm64	x64 Arm64
Windows	x64 x86 Arm64 winget instructions	x64 x86 Arm64
All	dotnet-install scripts	

Included runtimes

- .NET Runtime 8.0.29
- ASP.NET Core Runtime 8.0.29
- .NET Desktop Runtime 8.0.29

Language support

- CP 12.0
- FP 8.0
- Visual Basic 16.9

SDK 8.0.129

OS	Installers	Binaries
Linux	Package manager instructions	x64 Arm64 Arm32 x64 Alpine Arm64 Alpine Arm32 Alpine
macOS	x64 Arm64	x64 Arm64

Run apps - Runtime

ASP.NET Core Runtime 8.0.29

The ASP.NET Core Runtime enables you to run existing web/server applications. **On Windows, we recommend installing the Hosting Bundle, which includes the .NET Runtime and IIS support.**

IIS runtime support (ASP.NET Core Module v2)

18.0.26176.29

OS	Installers	Binaries
Linux	Package manager instructions	x64 Arm64 Arm32 x64 Alpine Arm64 Alpine Arm32 Alpine
macOS		x64 Arm64
Windows	x64 x86 Arm64 Hosting Bundle winget instructions	x64 x86 Arm64

.NET Desktop Runtime 8.0.29

The .NET Desktop Runtime enables you to run existing Windows desktop applications. This release includes the .NET Runtime; you don't need to install it separately.

OS	Installers	Binaries
Windows	x64 x86 Arm64 winget instructions	x64 x86 Arm64

.NET Runtime 8.0.29

The .NET Runtime contains just the components needed to run a console app. Typically, you'd



Two options

*Mac users

Texas A&M students have free access to SolidWorks Student Premium through the Software Center. **Recommended**

<https://software.tamu.edu/>

DO NOT
INSTALL IN
ONEDRIVE

DO NOT
INSTALL IN
ONEDRIVE

The software can also be accessed through the Virtual Open Access Lab (VOAL) which can be used without downloading SolidWorks.

For Mac users [Omissa
Horizon Clients]

and

People who are currently
installing SolidWorks

<https://voal.tamu.edu/>

Note: SolidWorks is the recommended software for TURTLE Advanced Projects. Please try to utilize SolidWorks, however, other CAD platforms may be accommodated.

VOAL Instructions



- Virtual Windows Machine that works anywhere
- Connect through TAMU VOAL site.
 - Login: Same as Howdy
 - {Engineering Desktop} click “...” then “Launch from Client”
- Do **NOT** save files in “(C:)” (You will lose the files)
 - Should be restricted by default
- Save Locations
 - “NetID (H:)” (Access only available in VOAL)
 - “Network or External Drive” (Need to grant VOAL access to the specific drive/folder)



Texas A&M University

<https://connect.voal.tamu.edu>

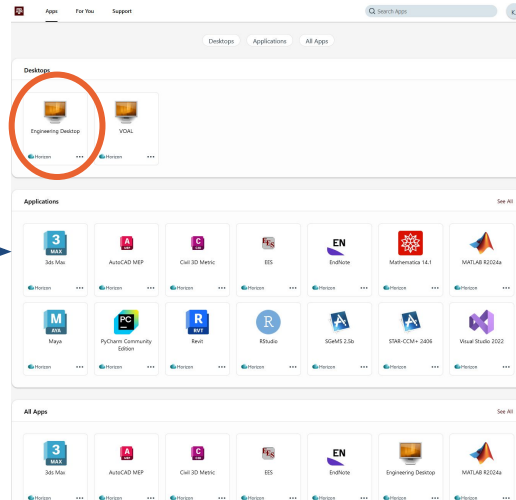
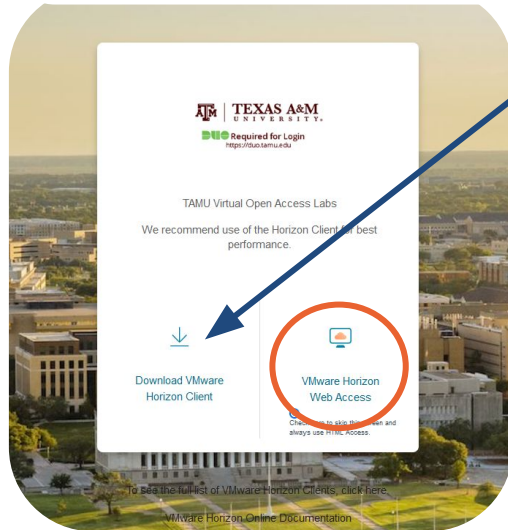
TAMU VOAL

TAMU Virtual Open Access Labs. We recommend use of the Horizon Client for best performance.

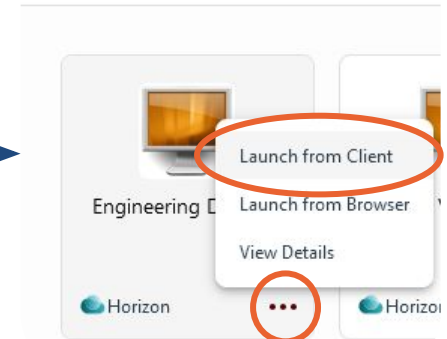
Download VMware Horizon Client. Launch Native Client.

Missing: ~~tau~~ | Show results with: ~~tau~~

Click the first time and download Omnissa Horizon Clients



Desktops



Bambu Labs Slicer



Slicer programs for 3D-Printing all function very similarly (Most are forks of Slic3r, Cura, or Simplify3D). This setup will depend on the brand, type of printers, and filaments available

Printer Selection

 Bambu Lab X1 Carbon ☒ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab X1 ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab X1E ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab P1P ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle
 Bambu Lab P1S ☒ 0.4mm nozzle ☐ 0.2mm nozzle ☒ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab A1 mini ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab A1 ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	

Most common filaments are:

1. PLA - Polylactic acid
2. PETG - Polyethylene terephthalate glycol

<https://bambulab.com/en-us/download/studio>

Software Links



- ANSYS - (Do **NOT** install in OneDrive)

<https://software.tamu.edu/>

- SolidWorks - (Do **NOT** install in OneDrive)

<https://software.tamu.edu/>

- Cisco Secure Client - (TAMU VPN)

https://connect.tamu.edu/+CSCOE+/logon.html?reason=12&gmsg=67686161727920726972656C6775766174#form_title_text

- “Omissa Horizon Client” - (TAMU VOAL Client)

https://customerconnect.omnissa.com/downloads/#all_products

- Bambu Lab Slicer -

<https://bambulab.com/en-us/download/studio>

- Printers: X1 Carbon 0.4 mm / P1S 0.4 and 0.6 mm
- Filaments: Generic PLA, Generic PETG, Bambu PLA Basic, Bambu PETG-HF



Pay your dues!!! Due on the 25th!!

Links should be in your orgs communication platform of choice (Discord)

Make sure you are in your orgs **Discord** and nicknamed “First Last” → Very important to not get kicked from the Discord
(Also turn on notifications for the announcements channel)



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Mechanical Design

Next Week



**“Your battles will be won with the help of
others”**—Kian’s Fortune Cookie

Attendance



Incubator

Lab Times

